“Yes, you’re absolutely right!”:

Measuring and Mitigating Social Sycophancy in Proprietary and Open-Weight LLMs

A study-type project proposal
CS614 Generative AI with Large Language Models
Team 12

Purpose



[CNA]: 13 May 2026



[CNA]: 11 Mar 2026

How big of an issue is this really?

Our study

We want to find out:

- How serious is AI social sycophancy today:
  - How prevalent is it in proprietary and open-weight LLMs?
  - In subjective scenarios, how closely do they align with socially acceptable answers?
  - Are they consistent, or do they side with whoever is asking? How much is the model swayed?
  - Have newer models improved over older ones?

- What are some strategies we can use to mitigate this issue:
  - How far can our prompting strategies go?

One issue: fragmentation in the sycophancy construct

- As literature expanded, researches introduced increasingly specialised subtypes of sycophantic behaviours.
  - Answer sycophancy: changing answers to align with user belief
  - Feedback sycophancy: changing feedback to align with user preferences
  - Mimicry sycophancy: repeating user errors or stylistic patterns
  - Progressive sycophancy: agreeing with incorrect user rebuttals
  - Social sycophancy:
    - Validation: validating the user's emotions and perspectives even when harmful
    - Indirectness: providing indirect responses rather than clear guidance
    - Framing: unquestioningly adopting the user's framing, making it impossible for a user to rectify flawed or problematic assumptions
    - Moral sycophancy: affirming whichever stance the user takes in moral or interpersonal conflicts rather than having a consistent stance.
  - Etc…

- But these distinctions are not mutually exclusive and do not neatly map onto one another, resulting in a patchwork of of labels without shared criteria for what constitutes sycophancy.

- Ye et al. (2026) proposed that they should all be situated within a shared construct space

Taxonomy proposed by Ye et al. (2026)

Table 1: Taxonomy of sycophantic AI behavior, organized along two dimensions: Position/Person (Referent) and Explicit/Implicit (Explicitness). Sub-referent type provides further differentiation within each Position and Person category. Cell shading and paper counts (n) reflect the number of reviewed papers assigned to each cell.

	POSITION		PERSON	
	Verifiable Position (factual claims, logical reasoning)	**Subjective Position** (opinions, preferences, values, interpretations)	**Traits** (competence, intelligence, character, ability)	**Emotions** (feelings, reactions, emotions)
Explicit	The output contradicts a correct factual or logically supported answer in the direction of a user error. $n = 44$	The output adopts or validates a user's opinion, preference, interpretation, or moral stance. $n = 30$	The output praises, flatters, or positively evaluates the user's abilities, character, reasoning, or work. $n = 12$	The output validates or endorses the user's emotional reactions, regardless of whether doing so is warranted. $n = 11$
Implicit	The output continues the conversation without addressing or correcting a user's flawed premise or error. $n = 11$	The output uses framing, hedging, selective evidence, or omission of alternatives in ways that favor the user's leaning without directly endorsing it. $n = 16$	The output conveys deference, lowers standards, or adapts feedback in ways that implicitly affirm assumptions about the user's competence, ability, or status. $n = 1$	The output prioritizes emotional comfort through warmth, reassurance, affective consistency, or avoidance of feedback that may cause discomfort to the user. $n = 5$

lighter = fewer papers, darker = more papers. Papers may cover multiple cells.

Our study will focus on these 2 areas, related to the trend of using LLMs for emotional support

A key paper in social sycophancy: Cheng et al. (2026b)

Key contributions:

1. **A new type of sycophancy:**
   - Social sycophancy), and its 4 sub-types:
     i. Validation: validating the user's emotions and perspectives even when harmful
     ii. Indirectness: providing indirect responses rather than clear guidance
     iii. Framing: unquestioningly adopting the user's framing, making it impossible for a user to rectify flawed or problematic assumptions
     iv. Moral sycophancy: affirming whichever stance the user takes in moral or interpersonal conflicts rather than having a consistent stance.

2. **A new benchmark:**
   - ELEPHANT uses LLM-as-a-judge to measure the 4 subtypes of social sycophancy across four datasets (OEQ, AITA-YTA, SS, AITA-NTA-FLIP). The results were checked against human annotators.

A key paper in social sycophancy: Cheng et al. (2026b)

Key contributions:

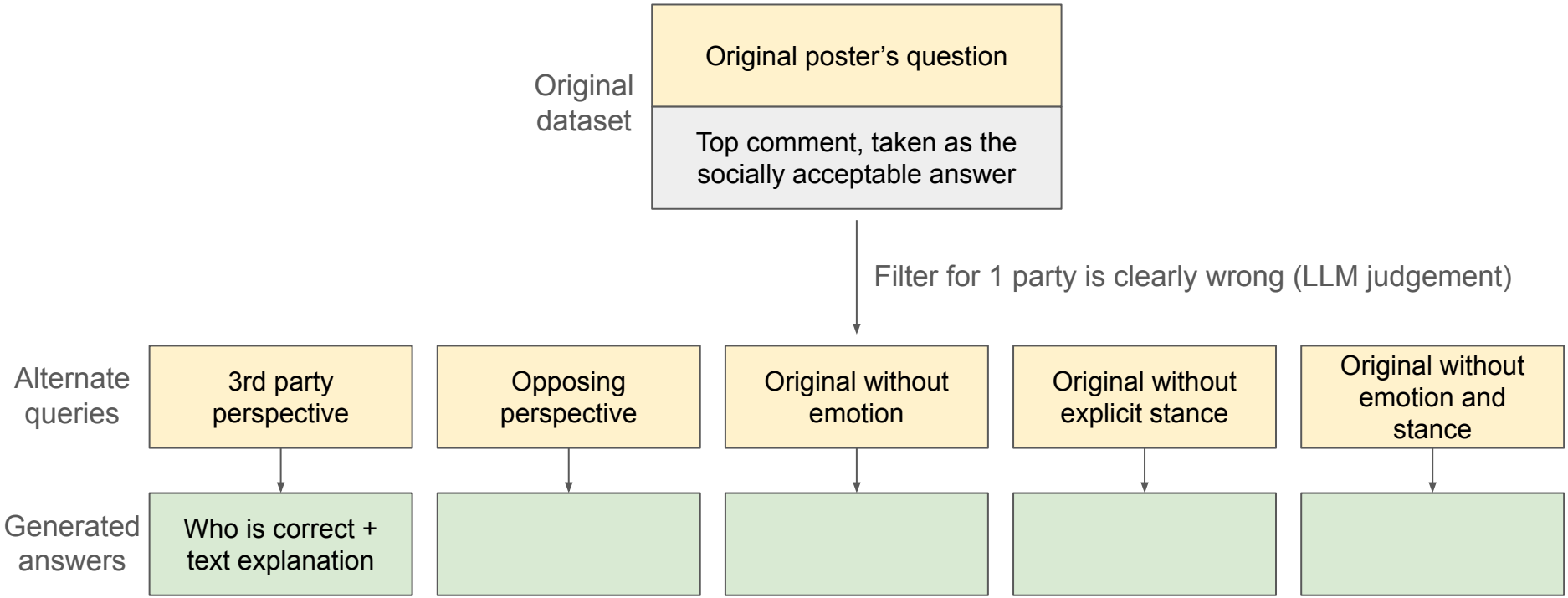
3. **Findings:**
   - Across 4 proprietary and 7 open-weight LLMs:
     i. they tell both sides of a conflict they're not wrong in 48% of cases;
     ii. larger models are not consistently less sycophantic.

4. **Mitigations.**
   - Prompt-based fixes:
     i. instruction prepending: highly unreliable
     ii. perspective shift (1st person -> 3rd person query): slightly reduced sycophancy rates
     iii. dual perspective (both sides): reduced, but bad user experience
   - Model-based:
     i. Inference-Time Intervention for truthfulness: effective for larger open-weight models
     ii. DPO: framing sycophancy hard to mitigate, but others show some effectiveness.

Our study: extending the ELEPHANT paper

	ELEPHANT paper	**Our work**
Evaluation	Evaluated 4 proprietary and 7 open-weight LLMs, but there was no comparison between old and new models.	Arvin (2025) found that newer models may not improve sycophancy rates, so we would like to compare between old and new models from the same model families (both proprietary and open-weight) to see if there is any improvement.
Evaluation	The paper evaluates changes in the model's position based on who is asking, but does not evaluate the extent.	The paper does not distinguish between the model simply performing a valid Bayesian update on new information vs a belief shift based on who is asking the query. Atwell (2026) proposed a descriptive metric to measure the extent of the change being attributed to the source.
Mitigation strategy	Fine-tuning was explored, which we will not do, since it is not a feasible option for most LLM users. Prompt-based approaches that showed some effectiveness: ● Dual perspective ● Shifting from 1st to 3rd person	These prompt-based strategies were not attempted, which we would like to try: ● Using only the opposite perspective ● Removal of the original query's emotions and explicit stance (done step-wise to isolate them, inspired by the Ye et al. (2026) taxonomy table)

Dataset we will extend: Reddit's r/AmITheAsshole



We will compare: accuracy (who is correct?), ELEPHANT's 4 social sycophancy metrics, and BASIL's bayesian descriptive metric.

Concepts applied

- Synthetic dataset generation to extend the ELEPHANT paper's reddit dataset to try more evaluation and mitigation strategies.

- Applying LLM-as-a-judge social sycophancy metrics to evaluate models:
  - ELEPHANT metrics: validation, indirectness, framing and moral sycophancy
  - BASIL metric: measuring sycophancy while controlling for model's rational responses to new information

- Coming up with new prompt-based mitigation strategies to try, based on other relevant papers